

Robert J. Noble

robjohnnoble.github.io
scholar.google.com/citations?user=IDDprHkAAAAJ

Research focus: Mathematical and computational modelling of cancer evolution and treatment.

Academic employment

- 2020- Department of Mathematics, City St George's, University of London
Senior Lecturer in Applied Mathematics (2024-)
Lecturer in Applied Mathematics (2020-24)
- 2018-2020 Department of Evolutionary Biology and Environmental Studies, University of Zurich
Postdoctoral researcher (20% contract) advised by Hanna Kokko
- 2017-2020 Department of Biosystems Science and Engineering, ETH Zurich
Postdoctoral researcher (80% contract from May 2018) advised by Niko Beerenwinkel
- 2014-2017 Institut des Sciences de l'Evolution de Montpellier (ISEM)
Postdoctoral researcher advised by Michael Hochberg

Education and qualifications

- 2024 Advance HE Fellowship
- 2009-2014 DPhil in Zoology (mathematical biology), University of Oxford
Supervisors: Mario Recker and Sunetra Gupta
- 1999-2003 Master of Mathematics (First Class), University of York

Publications and preprints (* denotes equal contributions)

- | | | |
|------|---|------------------------------------|
| 2026 | <i>Fluctuating DNA methylation sites encode colorectal tumour growth history</i>
Manojlović V, Gabbutt C, Shibata D, Noble RJ | bioRxiv
2026.06.04.730217 |
| 2026 | <i>When effective anticancer therapies are, in fact, destabilizing the tumor's Group Phenotypic Composition</i>
Thomas F, Dujon AM, Marusyk A, ..., Noble R , Nedelcu A, Gatenby R | npj Precision
Oncology |
| 2026 | <i>Selective sweep probabilities in spatially expanding populations</i>
Stein A, Bostock K, Kizhuttill R, Bak M, Noble RJ | Nature Commun.
17, 2181 |
| 2026 | <i>Preventing evolutionary rescue in cancer using two-strike therapy</i>
Patil S, Ahmed A, Viossat Y, Noble R | Genetics
232, iyaf255 |
| 2025 | <i>Evolutionary tree balance predicts disease-free survival in the TRACERx non-small cell lung cancer cohort</i>
Verity K, Noble R | medRxiv
2025.11.22.253407
97 |
| 2025 | <i>Expected and minimal values of a universal tree balance index</i>
Manojlović V*, Ahmed A*, Viossat Y, Noble R | arXiv
2507.08615 |
| 2024 | <i>A seven-step guide to spatial, agent-based modelling of tumour evolution</i>
Colyer B, Bak M, Basanta D, Noble R | Evol. Appl.
eva.13687 |
| 2023 | <i>A new universal system of tree shape indices</i>
Noble R , Verity K | bioRxiv
2023.07.17.549219 |
| 2023 | <i>Warlock: an automated computational workflow for simulating spatially structured tumour evolution</i>
Bak M, Colyer B, Manojlović V, Noble R | arXiv
2301.07808 |
| 2023 | <i>A survey of open questions in adaptive therapy: bridging mathematics and clinical translation</i>
West J, Adler F, Gallaher J, ..., Noble R , Viossat Y, Basanta D, Anderson ARA | eLife
12:e84263 |
| 2022 | <i>Robust, universal tree balance indices</i>
Lemant J, Le Sueur C, Manojlović V, Noble RJ | Syst. Biol.
71, 1210-24 |
| 2021 | <i>Spatial structure governs the mode of tumour evolution</i>
Noble R , Burri D, Le Sueur C, Lemant J, Viossat Y, Kather JN, Beerenwinkel N | Nature Ecol. Evol.
6, 207-17 |

- 2021 *Drug-induced resistance evolution necessitates less aggressive treatment* PLoS Comput. Biol. 17: e1009418
Kuosmanen T, Cairns J, **Noble R**, Beerenwinkel N, Mononen T, Mustonen V
- 2021 *Inferring the dynamic of mutated hematopoietic stem and progenitor cells induced by IFN α in myeloproliferative neoplasms* Blood 138, 2231-43
Mosca M*, Hermange G*, Tisserand A*, **Noble R***, ..., Plo I
- 2021 *Paracrine behaviors arbitrate parasite-like interactions between tumor subclones* Front. Ecol. Evol. 9:675638
Noble R, Walther V, Roumestand C, Hochberg ME, Hibner U, Lassus P
- 2021 *A theoretical analysis of tumour containment* Nature Ecol. Evol. 5, 826-35
Viossat Y, **Noble R**
- 2020 *Identifying key questions in the ecology and evolution of cancer* Evol. Appl. eva.13190
Dujon A, ..., **Noble R**, ..., Thomas F, Ujvari B
- 2020 *When, why and how tumour clonal diversity predicts survival* Evol. Appl. eva.13057
Noble R*, Burley JT*, Le Sueur C, Hochberg ME
- 2017 *Spatial competition constrains resistance to targeted cancer therapy* Nature Commun. 8, 1995
Bacevic K*, **Noble R***, ..., Hochberg ME, Krasinska L, Fisher D
- 2017 *Antibiotic stress selects against cooperation in the pathogenic bacterium Pseudomonas aeruginosa* PNAS 114, 546-51
Vasse M*, **Noble R***, ..., Hochberg ME
- 2017 *A framework for how environment contributes to cancer risk* Ecol. Lett. 20, 117-34
Hochberg ME, **Noble R**
- 2016 *Overestimating the role of environment in cancers* Cancer Prev. Res. 9, 773-6
Noble R, Kaltz O, Nunney L, Hochberg ME
- 2016 *A hypothesis to explain cancers in confined colonies of naked mole rats* bioRxiv 10.1101/079012
Hochberg ME, **Noble RJ**, Braude S
- 2015 *Peto's paradox and human cancers* Phil. Trans. B 370, 20150104
Noble R, Kaltz O, Hochberg ME
- 2013 *The antigenic switching network of Plasmodium falciparum and its implications for the immuno-epidemiology of malaria* eLife 2013.2:e01074
Noble R*, Christodoulou Z*, Pinches R, Kyes S, Recker M, Newbold CI
- 2012 *Erasing the epigenetic memory and beginning to switch—the onset of antigenic switching of var genes in Plasmodium falciparum* PLoS ONE 7, e34168
Fastman Y, **Noble R**, Recker M, Dzikowski R
- 2012 *A statistically rigorous method for determining antigenic switching networks* PLoS ONE 7, e39335
Noble R, Recker M

Software

- 2022- *warlock: Automated computational workflow for simulating tumour evolution* GitHub
- 2017- *ggmuller: Create Muller plots of evolutionary dynamics* CRAN

Teaching

- 2020- **Supervision (City St George's, University of London)**
PhD primary supervisor: Kate Bostock (2024-); Alexandru Chitiga (2024-); Kimberley Verity (2022-26); Blair Colyer (2021-25); Veselin Manojlović (2020-24)
PhD second supervisor: Manjot Singh Bedi (2023-); Youssef Arafat (2021-25); Hasan Haq (2021-25)
Internal master's project supervisor: Uliana Makhno (2026); Kenan Cetin Smith (2026)
External master's project supervisor: Srishti Patil (IISER, 2022-23); Carissa Mayo (PNRI, 2024-26)
Postdoc supervisor: Maciej Bak (2022)
- Lecturing and tutoring as module leader (City St George's, University of London)**
Mathematical processes for finance (BSc); Mathematics for economists post A Level (BSc); Forecasting (MSc)

Tutoring (City St George's, University of London)

Algebra (BSc); Functions, vectors and calculus (BSc); Final-year group projects (BSc)

2017- Supervision (ETH Zurich)

2020 Master's project supervisor: Alexander Stein; Jeanne Lemant; Dominik Burri

Research internship (eight months): Cécile Le Sueur

Lecturing and tutoring (ETH Zurich)

Evolutionary dynamics (MSc)

2016 Supervision (ISEM)

First year MEME MSc project: John Burley

2010- Tutoring & demonstrating (University of Oxford)

2013 Quantitative methods (BSc); Epidemiology (BSc); Epidemiological models (MSc)

Funding and awards

Co-awardee of US NSF grant *Quantifying and modeling the transmission dynamics of bivalve transmissible neoplasia* (co-supervised graduate student in 2023-27)

Sub-awardee of US NCI Arizona Cancer Evolution Center U54 grant (£110K in 2020-25)

Awardee of LMS Research in Pairs grant (£850 in 2024-26)

Awardee of Arizona Cancer Evolution Center Pilot Project funding (£6K in 2022)

Awardee of City University Pump Priming funding (£10K in 2022)

Co-awardee of FMJH grant *Optimization of a new type of cancer therapy* (£4K in 2019-20)

Biotechnology and Biological Sciences Research Council PhD fellowship (2009-13)

Travel grants: Lorentz Center (2017); Moffitt Cancer Center (2015); ECMTB (2011)

City University Images of Research Competition 2021: First prize

City St George's EDI/Civic Champion of the Year 2026: Runner Up

External professional activities

Associate Editor of the Journal of Evolutionary Biology (2025-); Guest Associate Editor of PLoS Computational Biology (2021-); Editorial Board Member of Scientific Reports (2024-25); Associate Editor of the Journal of Molecular Evolution (2022-25)

Vice-President (elected 2025) of the International Society for Evolution, Ecology and Cancer; previously elected Secretary (2022-25) and Advisory Committee member (2018-22)

Supervisory Board member of the EvoGamesPlus Innovative Training Network (2021-25)

PhD external examiner: University of Oxford (2025); Queen Mary University of London (2022, 2025), Institute of Cancer Research (2021)

PhD advisory committee member: University of Montpellier, ETH Zurich, Université Grenoble Alpes

Co-organizer: ISEEC Conference (2026); "Evolutionary approaches to understand cancer across scales" symposium (SMBE 2023); "Cancer Adaptive Therapy Models" workshop (2020); "Aging & cancer through the lens of evolution" symposium (ESEB 2019); "How does spatial structure affect tumour evolution?" symposium (MBE 2017)

Peer review for journals: American Naturalist, Bulletin of Mathematical Biology, Cancer Cell International, Cancer Research, Computational and Systems Oncology, eLife, Evolution, Evolutionary Applications, F1000Research, Frontiers Ecology and Evolution, Journal of Evolutionary Biology, Journal of Theoretical Biology, Mathematical Biosciences, Nature Communications, Nature Ecology & Evolution, Nature Genetics, Nature Reviews Bioengineering, npj Genomic Medicine, PLoS Evolutionary Biology, PLoS Computational Biology, PNAS, Proceedings of the Royal Society B, Royal Society Open Science, Scientific Reports

Peer review for funding agencies: French National Research Agency, Pancreatic Cancer UK

Other employment

- 2008-09 International HIV/AIDS Alliance (now known as Frontline AIDS): Communications
 2004-08 AVERT (HIV/AIDS charity): Science and health communication

Invited seminar talks

- Mar 2026 *Using mathematical modelling to support trials of evolutionary cancer treatment strategies*
 University of Bordeaux (virtual, hosted by Thomas Pradeu)
- Jan 2026 *Understanding, predicting and controlling stochastic cancer evolution*
 University of Edinburgh (hosted by Tibor Antal)
- Nov 2025 *Using mathematical modelling to support trials of evolutionary cancer treatment strategies*
 Institut de Recherche en Cancérologie de Montpellier (hosted by Daniel Fisher)
- Sep 2025 *Understanding, predicting and controlling stochastic cancer evolution*
 University of Oxford (hosted by Bridget Penman)
- Mar 2025 *Understanding, predicting and controlling stochastic cancer evolution*
 Mathematical and Computational Modeling of Cancer seminar (virtual, hosted by Ivana Bozic)
- Dec 2024 *Understanding, predicting and controlling stochastic cancer evolution*
 University of Sheffield (hosted by Alex Fletcher)
- Feb 2024 *New directions in mathematical oncology*
 Queen Mary University of London (hosted by Weini Huang)
- Apr 2023 *Quantifying and explaining modes of evolution*
 University of Leeds (hosted by Tyler Cassidy)
- Feb 2022 *Explaining the modes of tumour evolution*
 University of Warwick (virtual, hosted by Simon Graham)
- Oct 2021 *Explaining the modes of tumour evolution*
 University of Basel (virtual, hosted by Dominik Burri)
- Sept 2020 *Characterizing and forecasting tumour evolution*
 Cancer Research UK Cambridge Institute (virtual, hosted by Florian Markowetz)
- Jun 2020 *Characterizing and forecasting tumour evolution*
 Virtual Seminar on Modeling Biocomplexity (hosted by Andreas Deutsch)
- May 2020 *Characterizing and forecasting tumour evolution*
 Moffitt Cancer Center (virtual, hosted by David Basanta)
- Jan 2020 *Cancer: evolution, ecology and bad luck*
 University of Bath (hosted by Ben Ashby)
- Dec 2019 *The logic of containing tumours*
 University of Oxford (hosted by Eamonn Gaffney)
- Sep 2019 *Cancer: evolution, ecology and bad luck*
 University of Southampton (hosted by Lindy Holden-Dye)
- Feb 2019 *Characterising the evolutionary modes of cancer and normal tissue*
 TU Dresden (hosted by Andreas Deutsch)
- Mar 2018 *Characterising the evolutionary modes of cancer and normal tissue*
 University of Basel (hosted by Richard Neher)
- Feb 2018 *The mode and predictability of intra-tumour evolution*
 Wellcome Sanger Institute (hosted by Iñigo Martincorena)
- Dec 2017 *The mode and predictability of intra-tumour evolution*
 Boston University (hosted by Kirill Korolev)
- Nov 2017 *Spatial constraints on intratumour evolution*
 Harvard University (hosted by Martin Novak)
- May 2017 *Models for understanding tumour evolution and improving cancer therapy*
 University of Edinburgh (hosted by Bartłomiej Waclaw)
- Mar 2017 *Evolution, ecology, and cancer risk: from naked mole rats to modern humans*
 Chalmers University (hosted by Philip Gerlee)

- Sep 2016 *Cancer: evolution, ecology and bad luck*
Harvard University (hosted by Martin Novak)
- Feb 2015 *Data-based modelling of tumour evolution*
Moffitt Cancer Center (hosted by Robert Gatenby)

Conference talks

- May 2026 *Preventing evolutionary rescue in cancer*
International Society for Evolution, Ecology and Cancer Conference, Hersonissos
- Oct 2025 *Using mathematical models to support clinical trials of evolutionary cancer therapies*
Plenary talk at the Mathematical Oncology meeting, St. Pete Beach
- Aug 2025 *Selective sweep probabilities in tumours and other range expansions*
European Society for Evolutionary Biology conference, Barcelona
- May 2023 *Tumour heterogeneity and survival of cancer patients*
Invited talk at French Cancer Society training session on tumour heterogeneity, Paris
- Mar 2023 *Evolutionary approaches to overcoming cancer cell plasticity*
Invited talk at the Systems Approaches Towards Cancer Cell Plasticity Symposium, London
- Sep 2022 *Robust, universal tree balance indices*
European Conference on Mathematical and Theoretical Biology, Heidelberg
- Jul 2022 *Parasite-like interactions between tumour subclones*
Mathematical Models in Ecology and Evolution, Reading
- Jul 2021 *The evolutionary logic of tumour containment*
International Society for Evolution, Medicine & Public Health conference (virtual)
- Jun 2021 *The evolutionary logic of tumour containment*
Evolution conference (virtual)
- Jun 2021 *Explaining modes of tumour evolution*
Society for Mathematical Biology conference (virtual)
- Dec 2020 *The logic of containing tumours*
Cancer Adaptive Therapy Models workshop (virtual)
- Oct 2020 *Characterizing and forecasting tumour evolution*
International Symposium on Mathematical and Computational Oncology (virtual)
- Aug 2020 *The logic of containing tumours*
Invited talk at the Society for Mathematical Biology conference (virtual)
- Aug 2019 *Spatial competition constrains resistance to targeted cancer therapy*
International Society for Evolution, Medicine & Public Health conference, Zurich
- Jul 2019 *Spatial structure governs the mode of tumour evolution*
Intelligent Systems for Molecular Biology / European Conference on Comp. Biology, Basel
- Jun 2019 *Spatial structure governs the mode of tumour evolution*
Modelling Ecology & Evolution Zurich seminar, Zurich
- Sep 2018 *Characterising the evolutionary modes of cancer and normal tissue*
Evolutionary Models of Structured Populations workshop, Plön
- Dec 2017 *Spatial competition constrains resistance to targeted cancer therapy*
International Society for Evolution, Ecology and Cancer Conference, Tempe
- Oct 2017 *Impact of tissue architecture on the nature and predictability of tumour evolution*
Satellite Symposium to the Louis-Jeantet Symposium, Geneva
- Sep 2017 *Impact of tissue architecture on the nature and predictability of tumour evolution*
Basel Computational Biology Conference, Basel
- Jul 2017 *Impact of tissue architecture on the nature and predictability of tumour evolution*
Intelligent Systems for Molecular Biology / European Conference on Comp. Biology, Prague
- Apr 2017 *Evolutionary ecology of senescence and cancer risk: from naked mole rats to modern humans*
Modelling Biological Evolution conference, Leicester
- Nov 2016 *Controlling drug resistance with adaptive therapy*
Invited talk at the second Modeling Tumour Evolution conference, Bielefeld
- Sep 2016 *Cancer: evolution, ecology and bad luck*
Invited talk at the first Modelling Tumour Evolution conference, Bielefeld

- Jul 2016 *Cancer risk: evolution, ecology and bad luck*
Joint Meeting of ESMTB & Society for Mathematical Biology, Nottingham
- Dec 2015 *Peto's paradox and human cancers*
Third International Biannual Evolution and Cancer Conference, San Francisco
- Sep 2015 *Modelling ecological interactions of cancer clones*
Cancer Evolution Through Space and Time workshop, Plön
- Apr 2015 *Eco-evolutionary models of tumour heterogeneity*
Invited talk at the Modelling Biological Evolution conference, Leicester
- June 2011 *Using iterative methods to determine an antigenic switching network in Plasmodium falciparum*
European Conference on Mathematical and Theoretical Biology, Krakow
- May 2011 *Determining the switch pathway of the var gene repertoire of Plasmodium falciparum*
Biology and Pathology of the Malaria Parasite, Heidelberg